Akhilesh Siddhanti

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SUMMARY

Senior Member of Technical Staff at Salesforce with 4.5 years of experience building **AI powered backend/full-stack systems and agents** that enhance developer productivity. Skilled in **RAG**, **LLMs**, **semantic search**, **transformers** (SBERT, BERTopic), and **vector databases**, applied to **billions of documents** for **code intelligence and automation**. Published **11 research papers** with **147 citations** and reviewed **10+ IEEE submissions**.

WORK EXPERIENCE (5 Years)

Senior Member of Technical Staff, Salesforce, San Francisco **Jan 2021 - Present**

- Autonomous Test Failure Resolution Agent: Designed and scaled an AI agent that autonomously diagnoses and resolves **10,000+ daily test failures**, using **Sentence-BERT** and **Vertex AI Vector Search** over **1B+ vectors** to achieve **50% automated resolution accuracy**, improving release velocity and reliability for **6,000+ engineers**.
- Conversational Test Detective Agent & Autonomous Test Failure Assignment: Built a conversational agent using **DBSCAN clustering**, **XGBoost**, and **LLM-powered explainability** to identify culprit PRs among **1,000+ PRs per batch** with **70% accuracy**, reducing **mean time to resolution by 20%** and saving **1000+ engineering hours annually**.
- Efficient Text Preprocessing: Used **LLMs to clean redundant stacktrace information** and retain useful signals, a **96% reduction in stacktraces (from 1B to 40M)** and **decreasing project costs by 90% points**.
- Natural Language Modeling: Designed and deployed a scalable recommendation service with **natural language features**, improving recall by 17% points and **reducing prediction time from 10 minutes to seconds**, enabling real-time, batch inference at scale and supporting automated daily retraining with rigorous testing.
- ML Feature Engineering: Developed **10+ ML features**, including AST-based signals, and built a **file-mapping pipeline** that addressed **20% of recurring test failures**, boosting model recall by **8% pts**.
- Data Insights: Built and owned **company-wide dashboards** for Test Information and Predictive Performance, including a **Test Debug View** that helps engineers search and analyze stacktraces, accelerating debugging and data-driven decision-making for 50% of test failures across Salesforce.
- Performance and Scale Improvements:
  - Reduced feature ingestion time from **2 weeks to 2 days**, boost model recall by **25% pts**.
  - Identified performance pitfalls by switching to batch prediction, **decreasing 97% in CPU usage**
  - Identified Test time-binning pitfalls to recover compute and **increase tests recommended by 55% pts**
  - Optimized feature pipeline by **decreasing memory footprint by 1000x** and **making it 4x faster**.
- Infrastructure and Trust work: Built and **owned company-wide dashboards** for test analytics and recall performance, including fixing 90% Flaky tests and 50% of Test Failures

Software Development Intern, Salesforce **May 2020 - Aug 2020**

- Performed migrations to a new Test Inventory service, **increasing reliability by 100%**

SKILLS

Languages: Java, Python, C/C++, JavaScript, SQL, BigQuery, Elasticsearch, Terraform, Pandas, NumPy
Tools/Frameworks: Springboot, Kafka, REST, PyTorch, TensorFlow, Scikit-learn, Hugging Face, XGBoost, FAISS, Vertex AI, Git, Docker, Jenkins, Kubernetes, CI/CD, AWS, GCP, Postman
Concepts: CV, NLP, ML, LLMs, RAG, Prompt Engineering, Embedding Models, Semantic Search, Clustering, Explainable AI, SHAP

PUBLICATIONS, in International Conferences and Journals (11)

Differential fault location identification by Machine Learning **CAAI Transactions on Intelligence Technology**
A Differential Fault Attack on Plantlet (Cipher) **IEEE Transactions on Computers**
+ 9 other international papers in Computer Science, Total Citation Count: 147

EDUCATION

MS in Computer Science, Georgia Institute of Technology, Atlanta **Aug 2019 - Dec 2020**
M.Sc. (Hons.) in Mathematics and B.E. (Hons.) in Computer Science, BITS Pilani **Aug 2014 - May 2019**